3.17 METALWORK (445)

Metalwork Paper

SECTION A marks

Answer all the questions in this section in the spaces provided

(a) Define the term "apprentice" as applied in the engineering field. (1 mark)

b Explain the term "break even" as used in business. mark

a List **three** uses of a steel rule. ($\frac{1}{2}$ marks)

b i With the aid of sketches distinguish between a dot punch and centre punch. marks

ii State **two** uses of a dot punch. mark

a State **two** reasons for edge treatment on sheet metal articles. marks

(b) Outline the procedure of finishing a work piece by painting. (2 marks)

a Outline the process of case hardening a vee block. ($\frac{1}{2}$ marks)

b State **two** effects of each of the following alloying elements on iron.

i chromium mark

ii manganese mark

(a) Define the term "upsetting" as used in forging. (1 mark)

b State **two** reasons for twisting metal bars. marks

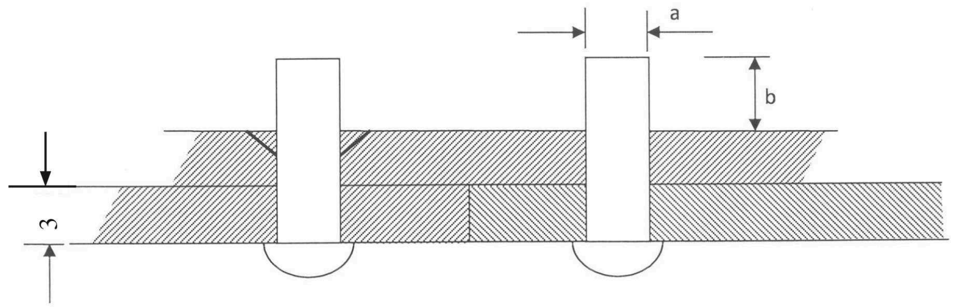
(a) With respect to needle files, state:

i their use mark

(ii) the reason for not fitting a handle; (1 mark)

iii the reason for knurling one end. mark

(b) **Figure □** shows two mild steel plates of equal thickness to be rivetted.



Determine:

(i) rivet diameter marked a;

(ii) heading allowance marked b.

(□ marks)

□. (a) State the:

(i) effect of prolonged heating in brazing; (□ mark)

(ii) reason for concentrating heat on the thicker piece of metal when brazing two metals. (□ mark)

(b) With reference to arc welding:

(i) define the term "tack welding"; $(1\frac{1}{2}$ marks)

(ii) state the use of tacks. $(\square\frac{1}{2}$ marks)

□. With the aid of labelled sketches, distinguish between parallel turning and facing in lathe work. (□ marks)

□. State **four** possible causes of burns in a workshop. (□ marks)

□. **Figure □** shows an isometric drawing of a block. Sketch in third angle projection the orthographic views of the block. (□ marks)

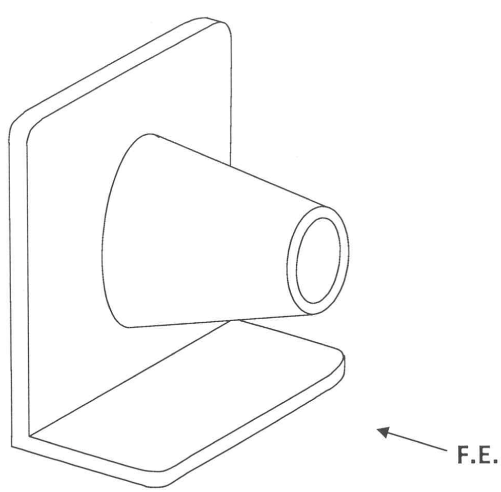


Figure 2

SECTION B marks

Answer question and any other three questions from this section

Candidates are advised to spend not more than minutes on question

Figure shows two views of a machined component drawn in first angle projection.

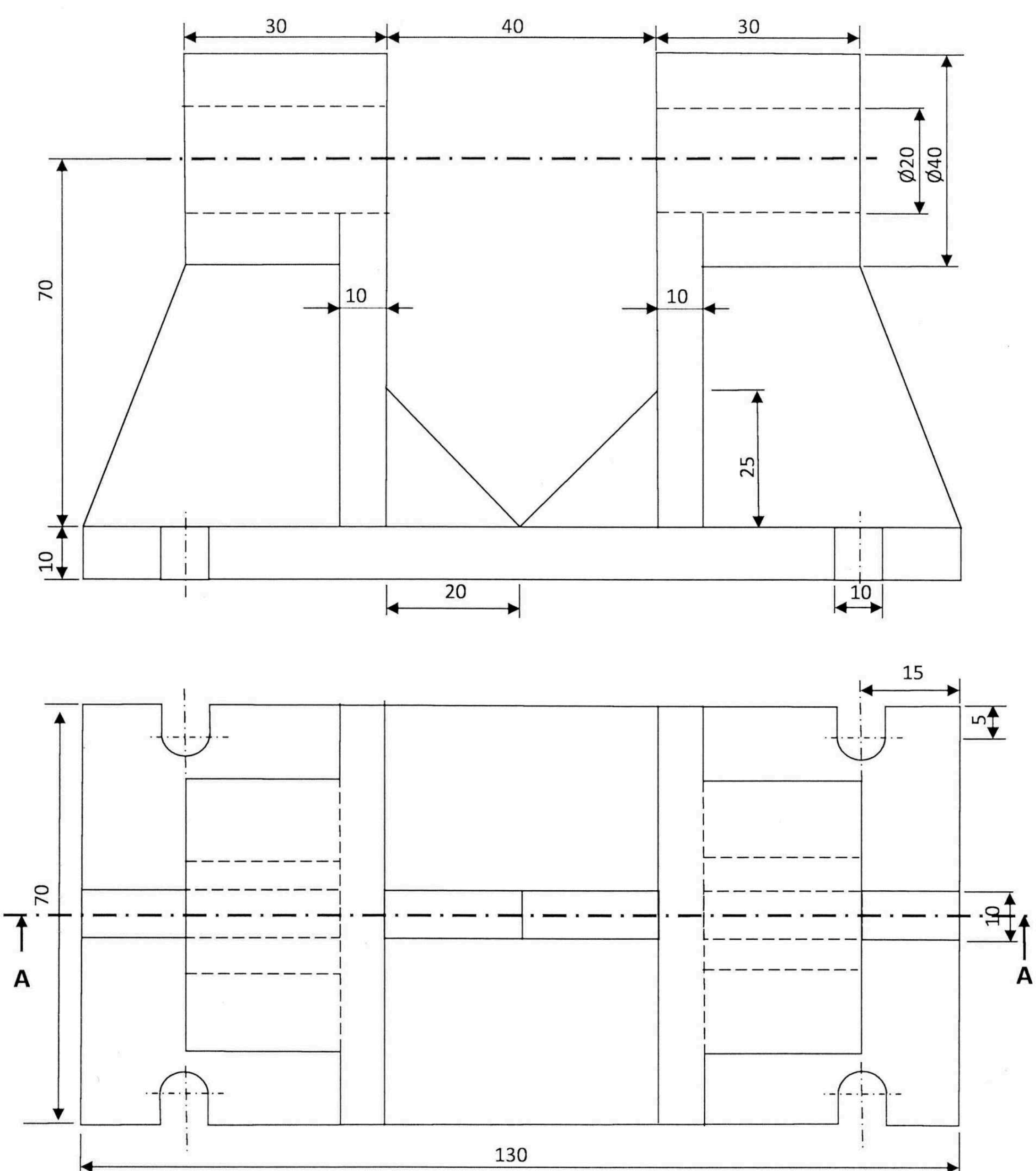


Figure 3

Draw full size, the following views:

sectional front elevation through A A

end elevation include hidden details

marks

(a) State **four** safety precautions to be observed before switching on power supply of an electric arc welding equipment. (marks)

(b) With respect to arc welding:

(i) state the **two** methods of striking the arc. (mark)

(ii) give **one** advantage and **one** disadvantage of using each method. (marks)

(c) Name and illustrate **four** welding defects. (marks)

(a) With the aid of sketches, explain the procedure of drilling a hole on a centre lathe machine. (marks)

(b) State and sketch **two** methods of producing a short taper on a lathe machine. (marks)

(a) Outline the procedure of cutting internal threads on a round bar using a die stock. (marks)

(b) Name and sketch the three thread taps which make a set. ($\frac{1}{2}$ marks)

(c) Sketch in pictorial a hand file and label all its parts. (5$\frac{1}{2}$ marks)

(a) State **two** causes for each of the following problems in drilling:

(i) worn out corners of cutting edges on a twist drill. (marks)

(ii) chipped cutting lips. (marks)

(iii) rough walls of a drilled hole. (marks)

(b) With the aid of labelled sketches show how:

(i) a centre punch is ground on a grinding wheel. (4$\frac{1}{2}$ marks)

(ii) the grinding lines should appear on the ground surface of the centre punch. ($\frac{1}{2}$ marks)

(c) State **three** safety precautions to be observed when grinding. (marks)